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Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S2 (R) Examination May 2025 (2024 Scheme)

Course Code: GXEST204

Course Name: PROGRAMMING IN C

Max. Marks: 60 Duration: 2 hours 30 minutes

PART A

(Answer all questions. Each question carries 3 marks) CO Marks

1 What will be the output of the following C program? Justify your answer. CO1 (3)

```
#include <stdio.h>
int main() {
  int a = 5; // binary: 0101
  int b = 9; // binary: 1001
  printf("%d", a & b);
  return 0;
}
```

2 Find out the errors, if any in the following program. Correct the error and justify your answer. CO1 (3)

```
Include  <math.h>
main  { }
(
FLOAT  X;
X=2.5;
Y=exp(x);
Print(x,y);
)
```

3 Explain enumerated data types with example. CO2 (3)

4 Write a C program to initialize a 3×3 matrix and display its diagonal elements separately. CO2 (3)

5 List the four storage classes in C and mention one feature of each. CO3 (3)

6	Explain with an example how a union saves memory compared to a structure.	CO3	(3)
7	What is the purpose of malloc() and free() functions?	CO4	(3)
8	How do you open a file for reading and writing in C? Write the syntax.	CO4	(3)

PART B

(Answer any one full question from each module, each question carries 9 marks)

Module -1

9	a)	Write a C Program to check if a given number is a strong number or not. A strong number is a number in which the sum of the factorial of the digits is equal to the number itself. Eg:- 145: 1 !+4 !+5!: 1 +24+120=145	CO1	5
	b)	Write a C program to display first 50 prime numbers.	CO1	4
10	a)	Write a C program to print sum of n natural numbers. Also count the number of odd numbers in it.	CO1	3
	b)	Explain any 3 bitwise operators and 3 logical operators in c with examples.	CO1	6

Module -2

11	a)	Write a C program that sorts an array of strings alphabetically using strcmp() and strcpy() functions.	CO2	6
	b)	What will be the output of the following C program? Justify your answer.	CO2	3

```
#include <stdio.h>
#include <string.h>
int main() {
    char str1[10] = "abc";
    char str2[10] = "def";
    strcat(str1, str2);
    printf("%s", str1);
    return 0;
}
```

12	a)	Write a C program that reads a string from the user, counts the number of vowels, consonants, digits, and special characters and display the counts separately.	CO2	5
	b)	Write a C program to insert a new element into an existing array at a given position, shift the remaining elements accordingly, and print the new array.	CO2	4
		Module -3		
13	a)	Write a C program that uses a function to sort an array of integers.	CO3	5
	b)	Write a C program to define a structure Book (title, author, price). Write a function to accept and display a book's details.	CO3	4
14	a)	Write a C program to find the factorial of a given number using recursion.	CO3	6
	b)	Explain the difference between recursion and iteration.	CO3	3
		Module -4		
15	a)	Write a C program to read an array of 5 integers. Use a pointer to find and print the largest element of the array.	CO4	4
	b)	Explain any 5 file handling functions in C?	CO4	5
16	a)	Write a C program to concatenate two strings using pointers.	CO4	5
	b)	Write a C program to read integers from a text file and count how many even and odd numbers are present.	CO4	4

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